CORRECTION NOTICE

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In the version of this supplementary file originally posted online, the legends for Supplementary Figures 2b, 3, 6, 7, 8 and 10 were inadvertently deleted. The legends have been provided in this file as of 7 June 2011.

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Supplementary Information

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Supplementary Results

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Supplementary Methods

Construction of the *frsA*-deleted mutant

An 838-bp DNA containing the *frsA* upstream region was amplified from the genomic DNA of *V. vulnificus* MO6–24/O[1] using two primers, frsA-upF (5'-ACATGCATGCAATAATCGTTTGCGCAGCTCGATACCC-3'; underlined sequence denotes an SphI restriction site) and frsA-upR (5' -GCTCTAGATCGGCATGTATTGAGTCAATGCCGAGG -3'; underlined sequence denotes an XbaI restriction site). The PCR product was then cloned into pBluescript SK II(+) to produce pSK*frsA*up. A 1,118-bp DNA fragment containing the downstream region of the *frsA* gene was made using the primers frsA-downF (5'-GCTCTAGAAGGGGATCCGGTCTCGCCATATTCGGA-3'; underlined sequence denotes an XbaI restriction site) and frsA-downR (5'-GGACTAGTATCCGCTCGAGTGAGCAACATTTGGCC-3'; underlined sequence denotes an SpeI restriction site) and cloned into the corresponding sites of pSK*frsA*up to result in pSK*frsA*up/down. A 1,956-bp DNA fragment of pSK*frsA*up/down digested with SphI and SpeI was ligated into a suicide vector, pDM4[2], to generate pDM4-Δ*frsA*. The *E. coli* SM10λ*pir* strain[3] carrying pDM4-Δ*frsA* was conjugated with *V. vulnificus* MO6–24/O, and the exconjugants were then selected on thiosulfate citrate bile sucrose medium as described previously[4]. The selected colony was further confirmed by PCR using primers frsA-upF and frsA-downR and Δ*frsA* mutant was named with SM201. For complementation of the mutant, a 1,731-bp DNA containing the *V. vulnificus frsA* gene, which had been amplified using two primers, frsA-comF (GCTCTAGAAACTGCGTGAAATGTACCCTAAAGCG-3'; underlined sequence denotes an XbaI restriction site) and frsA-comR (5'-

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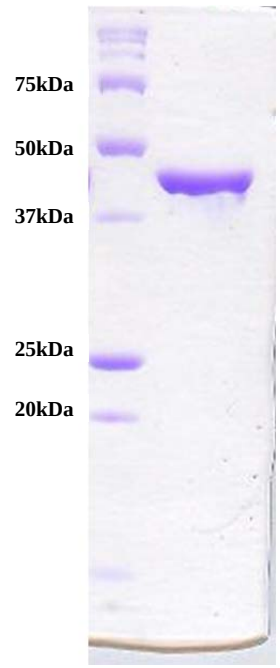
GGGGTACCTTGCCAATAAACGATTGTGGGTAGGG-3'; underlined sequence denotes a KpnI restriction site), was cloned into the broad-host-range plasmid pRK415. The resultant plasmid, pRK415-*frsA*, was transformed into *E. coli* SM10 λ*pir* and then transferred to SM201 by conjugation.

Cloning and purification of FrsA

Two oligonucleotides, FrsAexp-FK (5'-GGGGTACCCCGAATATGTCAGAAGAAGTCAGC-3'; underlined sequence denotes a KpnI restriction site) and FrsAexp-RH (5'-CCCAAGCTTGTCACCTTAAGAGTTCATCTTCCAGC-3'; underlined sequence denotes a HindIII restriction site), were used to amplify a 1,255-bp DNA fragment containing the complete open reading frame of the *frsA* gene from *V. vulnificus* genomic DNA. KpnI and HindIII sites located at both ends of the resultant *frsA* DNA were used to clone this DNA into the pQE30 expression plasmid (Qiagen), to generate a plasmid pQE-*frsA*. Recombinant FrsA was overexpressed in *E. coli* JM109 (Promega) by adding isopropylthio-β-*D*-galactoside at a concentration of 0.2 mM, and purified using Ni^{+}-nitrilotriacetic acid affinity column and size exclusion chromatography (**Supplementary Fig. 1**).

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Supplementary Fig. 1. SDS-PAGE of the purified recombinant FrsA protein. A polyacrylamide gel run by the purified recombinant FrsA protein and subsequently stained by Coomassie Brilliant Blue R showed the high purity of the recombinant protein used in this study.



Cloning and purification of IIA^{Glc}

Two oligonucleotides, Crrexp-F (5'- CGGGATCCGACACAATGGGTCTGTTTGAC -3'; underlined sequence denotes a BamHI restriction site) and Crrexp-R (5'-AACTGCAGTAGTAATTACTTAGTTACGCG -3'; underlined sequence denotes a PstI restriction site), were used to amplify a 522-bp DNA fragment containing the complete open reading frame of the *crr* gene encoding IIA^{Glc} from the genomic DNA of *V. vulnificus*. BamHI and PstI sites located at both ends of the resultant *crr* DNA were used to generate a plasmid pQE-*crr*. Recombinant phosphorylated IIA^{Glc} or dephosphorylated IIA^{Glc} were overexpressed in *E. coli* JM109 grown in the medium containing 1.0 mM phosphoenolpyruvate or 1.0% glucose, respectively, and purified using an Ni^{+}-nitrilotriacetic

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acid affinity column and size exclusion chromatography.

Site-directed mutagenesis of FrsA

Amino acid residues were mutagenized by overlap extension[5] using the primers that include the altered nucleotides to substitute alanine for the 53rd arginine, the 203rd aspartic acid, or the 272nd arginine. To amplify the DNA encoding R53A FrsA, two sets of primers, FrsAexp-FK/ FrsA R53AR (5'-CCACACCCATTGCAAGCGAGCAAGGTTGCG-3') and FrsAexp-RH /FrsA R53AF (5'-CGCAACCTTGCTCGCTTGCAATGGGTGTGG-3'), were utilized. Two PCR products were then used as templates for the next PCR using FrsAexp-FK and FrsAexp-RH. The resultant mutagenized *frsA* DNA was digested with KpnI and HindIII, and ligated to KpnI/HindIII-digested pQE30 to produce pQE-*frsA*R53A. Similarly, other sets of primers, FrsA D203AR (5'-GGTTTGCAGACTGGCTAATCCCGCACTCAC-3')/ FrsA D203AF (5'-GTGAGTGCGGGATTAGCCAGTCTGCAAACC-3'), FrsA D203NR (5'-GGTTTGCAGACTGTTTAATCCCGCACTCAC-3')/ FrsA D203NF (5'-GTGAGTGCGGGATTAAACAGTCTGCAAACC-3'), and FrsA R272AR (5'-GTTACCGCCAAAAGCAAAGCCGATCAAACC-3')/ FrsA R272AF (5'-GGTTTGATCGGCTTTGCTTTTGGCGGTAAC-3'), were used to produce pQE-*frsA*D203A, pQE-*frsA*D203N, and pQE-*frsA*R272A, respectively. For double mutation of R53A and R272A, the primer set, FrsA R272AR/FrsA R272AF, was utilized to amplify the DNA using pQE-*frsA*R53A as a template to produce pQE-*frsA*R53A/R272A. Nucleotide sequences of all the resultant plasmids were confirmed by DNA sequencing and then mobilized into *E. coli* JM109 to purify the recombinant proteins as described above.

Estimation of enzyme kinetic parameters

FrsA enzyme assay was performed in 50mM sodium phosphate solutions (pH 7.0) containing

1 mM dithiothreitol and various concentrations of pyruvate. To measure acetaldehyde produced from pyruvate, 2 mM NADH and 3.7 U of yeast alcohol dehydrogenase (ADH) (Sigma Co.) were included in the reaction mixtures[6]. Reaction was initiated by adding the recombinant FrsA (1 nM) and the reaction mixtures were incubated at 37 ^{o}C for 1 min. The remaining NADH in the reaction mixtures was continuously determined by spectrophotometry at 340 nm. To measure CO_2, the enzyme reaction was performed in 50 mM Tris-HCl buffers (pH 7.0). Carbon dioxide dissolved in the reaction solutions was quantified by a spectrophotometry after the reaction solutions were treated with 10 mM calcium hydroxide[7]. Concentrations of pyruvate and acetaldehyde in cell lysates or reaction mixtures were measured by the pyruvate assay kit (BioVision) and the acetaldehyde UV-method kit (Roche), respectively. To increase the sensitivity of the acetaldehyde UV-method kit, the reaction mixtures were further treated with a color reagent (40 mg iodonitrotetrazolium chloride and 10 mg phenazene methosulfate in 50ml of 1.0% Tween 20) and then spectrophotometrically measured at 490 nm[8].

NMR spectroscopy

Five millimolar pyruvate, α-ketoglutarate, or α-ketobutyrate was added to a solution (50 mM sodium phosphate and 1 mM dithiothreitol in 95% D_2O). Reaction mixtures were incubated at 20 ^{o}C for 90 min in the absence or presence of 2 μM FrsA. After keeping reaction mixtures on ice for 2 hrs, they were subjected to the NMR experiments. The spectra were recorded at a ^{1}H resonance frequency of 500 MHz by use of a Bruker DRX500 spectrometer.

Mouse mortality tests

All mice were housed in specific pathogen-free conditions at Yonsei University College of Medicine, and all animal experiments were performed in accordance with Institutional

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Animal Care and Use Committee guidelines. For determination of LD_{50}, specific pathogen-free, 7-week-old, female ICR mice were used without pretreatment with iron dextran. Bacterial cultures grown in LBS medium at A_{600} of 1.0 were harvested, washed in PBS, and then resuspended in PBS. One hundred microliters of serial dilutions of the bacterial suspension were injected intraperitoneally into six mice per each dilution group. The numbers of dead mice were determined at 24 h after the injection, and the LD_{50} values (colony-forming units: CFU) were calculated as described previously[9].

Crystallization, structure determination and refinement of FrsA

Crystals were obtained at 23 °C by the batch crystallization method with mother liquors of 1.8 M ammonium sulfate, 0.1 M Tris-HCl (pH 8.5), 0.2 M sodium acetate, and 5 mM oxamate. The crystals belonged to the space group $P2_1$ with cell parameters a = 60.834, b = 88.201, c = 79.629 Å and β = 106.619º. For data collection, crystals were frozen at -170 °C after a brief immersion in a cryoprotectant solution containing 15% ethylene glycol in the same mother liquor. A 2.2 Å data set (**Supplementary Table 2**) was collected at beamline 17A at Photon Factory, Japan.

The structure was solved by the MAD method using three-wavelength data sets of SeMet-labeled FrsA (**Supplementary Table 2**). The 26 selenium positions of crystals of SeMet-labeled FrsA were located, and phases were calculated using the program Solve. The subsequent solvent flattening by Resolve gave rise to an interpretable map, based on which the *de novo* model building of FrsA was easily completed. The model building was performed using Quanta and refinement was done with maximum likelihood algorithm implemented in CNS (**Supplementary Table 2**).

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Model building of the FrsA/puruvate complex

It is well established that the oxyanion hole that exists commonly in α/β hydrolase structures accommodates the carbonyl oxygen of substrates in ester or peptide hydrolysis reactions[10]. Based upon the "solid docking" module in QUANTA (Molecular Simulation, Inc.), pyruvate is positioned to locate its carbonyl oxygen between the backbone -NH groups of Leu202 and Phe273 that correspond to the oxyanion hole in α/β hydrolases. The module allows a successful docking only when an electrostatic and geometric complementarity is accomplished between a host and a guest molecule.

Crystallization, structure determination and refinement of the FrsA/d-IIA^{Glc} complex

Purified FrsA and IIA^{Glc} proteins were mixed with 1:1 molar ratio and the mixture was used for crystallization. Crystals were obtained at 23 °C by the batch crystallization method with mother liquors of 20 % polyethylene glycol monomethyl ether 550, 0.1 M MES (pH 6.5), and 0.01 M zinc acetate. The crystals belonged to the space group *C*2 with cell parameters a = 322.87, b = 62.07, c = 126.36 Å and β = 110.8º. For data collection, crystals were frozen at -170 °C after a brief immersion in a cryoprotectant solution containing 15% ethylene glycol in the same mother liquor. A 2.2 Å data set (**Supplementary Table 3**) was collected at beamline 17A at Photon Factory, Japan.

The structure was solved by the molecular replacement method using MolRep in the *CCP*4 program suite. The crystal structures of FrsA and IIA^{Glc} (PDB ID; 1O2F) was used as search models. The model building was performed using Quanta and refinement was done with maximum likelihood algorithm implemented in CNS (**Supplementary Table 3**).

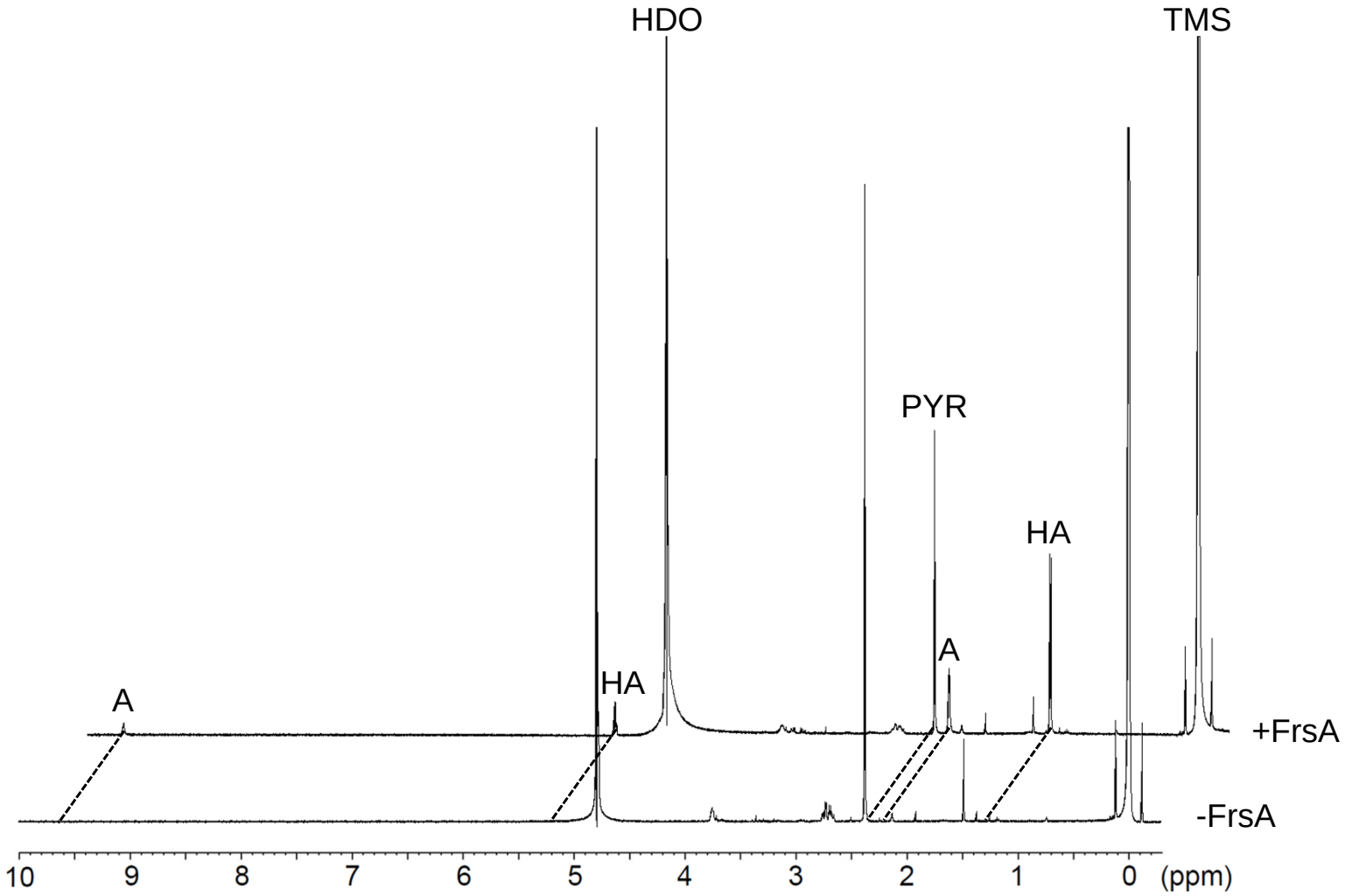
REFERENCES

1. Wright, A.C., Simpson, L.M., Oliver, J.D. & Morris, J.G., Jr. Phenotypic evaluation of acapsular

transposon mutants of *Vibrio vulnificus*. *Infect Immun* **58**, 1769-73 (1990).
2. Milton, D.L., O'Toole, R., Horstedt, P. & Wolf-Watz, H. Flagellin A is essential for the virulence of *Vibrio anguillarum*. *J Bacteriol* **178**, 1310-9 (1996).
3. Simon, R., Priefer, U. & Puhler, A. A broad host range mobilization system for *in vivo* genetic engineering: transposon mutagenesis in Gram negative bacteria. *Nat Biotechnol* **1**, 784-791 (1983).
4. Park, K.J. et al. Isolation and characterization of *rpoS* from a pathogenic bacterium, *Vibrio vulnificus*: role of σ^S in survival of exponential-phase cells under oxidative stress. *J Bacteriol* **186**, 3304-12 (2004).
5. Sambrook, J. & Russell, D.W. *Molecular cloning: a laboratory manual*, (Cold Spring Harbor Laboratory Press, Cold Spring Harbor, NY, 2001).
6. Ullrich, J. Yeast pyruvate decarboxylase (2-oxoacid acrboxylase, EC 4.1.1.1) assay of thiamine pyrophosphate. *Methods Enzymol.* **18**, 109-115 (1970).
7. Gu, W., Bousfield, D.W. & Tripp, C.P. Formation of calcium carbonate particles by direct contact of $Ca(OH)_2$ powders with supercritical CO_2. *J Mater Chem* **16**, 3312-3317 (2006).
8. Hinman, L.M. & Blass, J.P. An NADH-linked spectrophotometric assay for pyruvate dehydrogenase complex in crude tissue homogenates. *J Biol Chem* **256**, 6583-6 (1981).
9. Reed, L.J. & Muench, H. A simple method of estimating fifty per cent endpoints. *Am. J. Epidemiol.* **27**, 493-497 (1938).
10. Nardini, M. & Dijkstra, B.W. α/β hydrolase fold enzymes: the family keeps growing. *Curr Opin Struct Biol* **9**, 732-7 (1999).

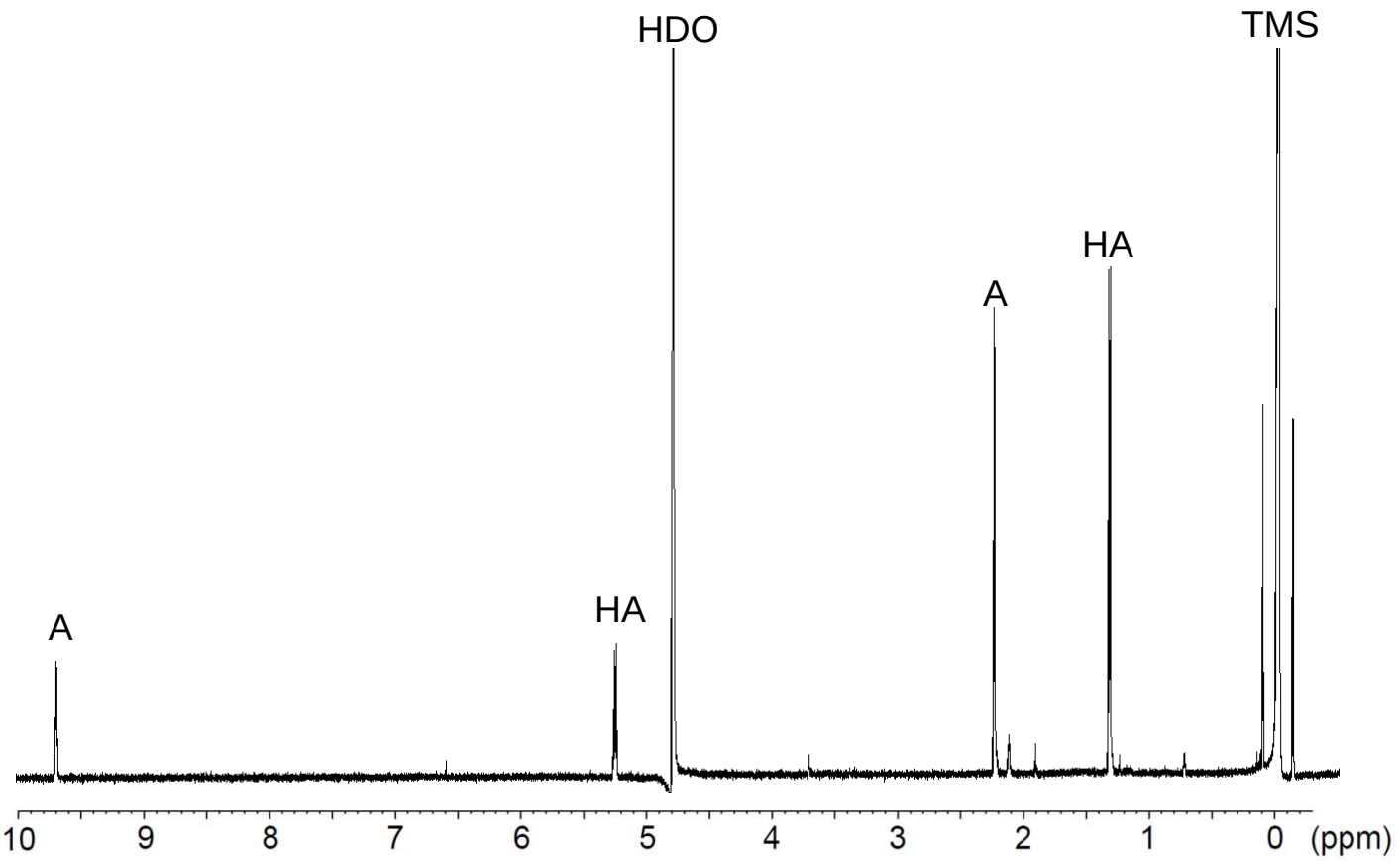
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Supplementary Results



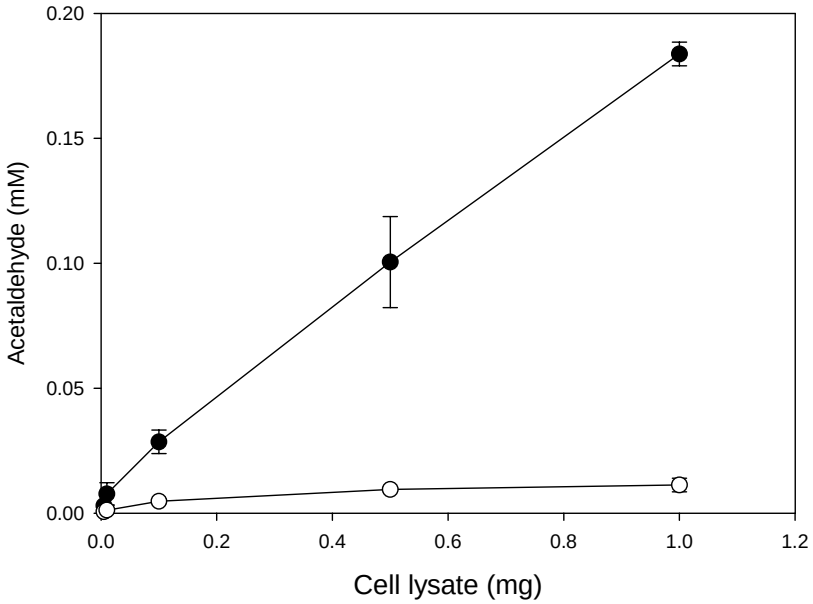
Supplementary Fig. 2a. Enzymatic activity of FrsA. Conversion of pyruvate to acetaldehyde by FrsA. Pyruvate (5 mM) was added to 50 mM sodium phosphate buffered D_2O and incubated in the absence or presence of 2 μM FrsA. The reaction mixtures were analyzed by ^{1}H-NMR as described in Supplementary Methods. The following abbreviations are used: A, acetaldehyde; AH, hydrated acetaldehyde; PYR, pyruvate; TMS, tetramethylsilane. A standard ^{1}H-NMR spectrum for acetaldehyde was presented in Supplementary Fig. 2b.

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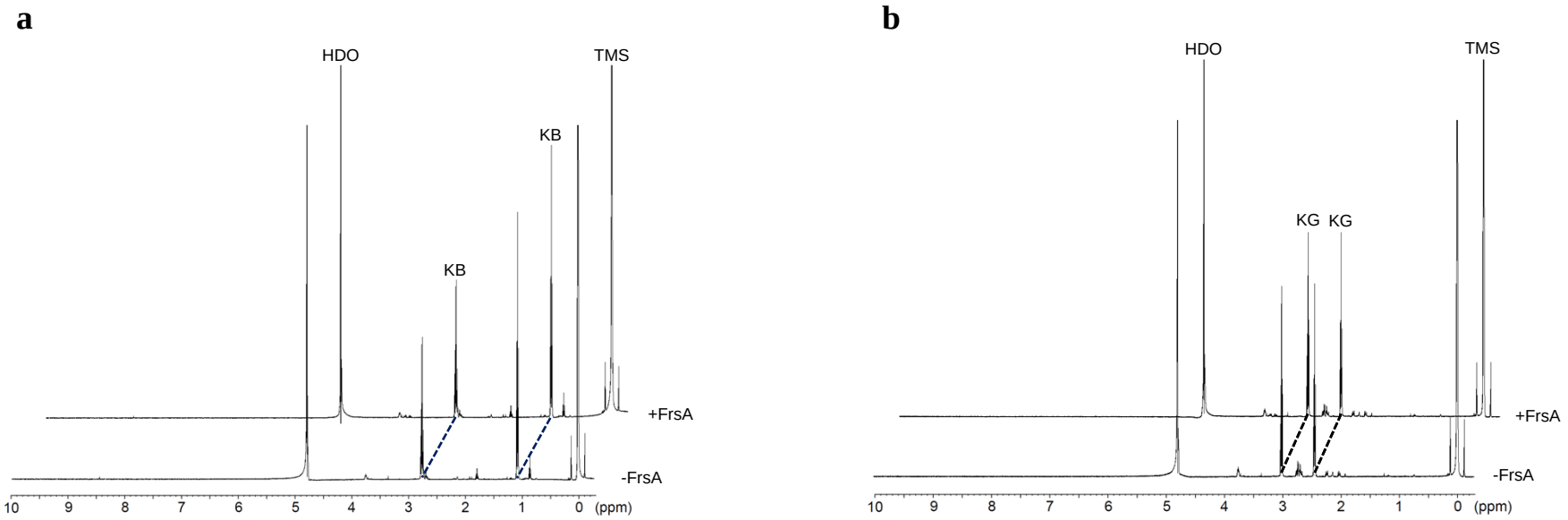
Supplementary Fig. 2b. 500MHz ^{1}H-NMR spectrum of the pure acetaldehyde. The peaks corresponding to acetaldehyde (purchased from Sigma Co.) are designated by A (acetaldehyde: at 2.24 and 9.68 ppm) and HA (hydrated acetaldehyde: at 1.33 and 5.25 ppm).

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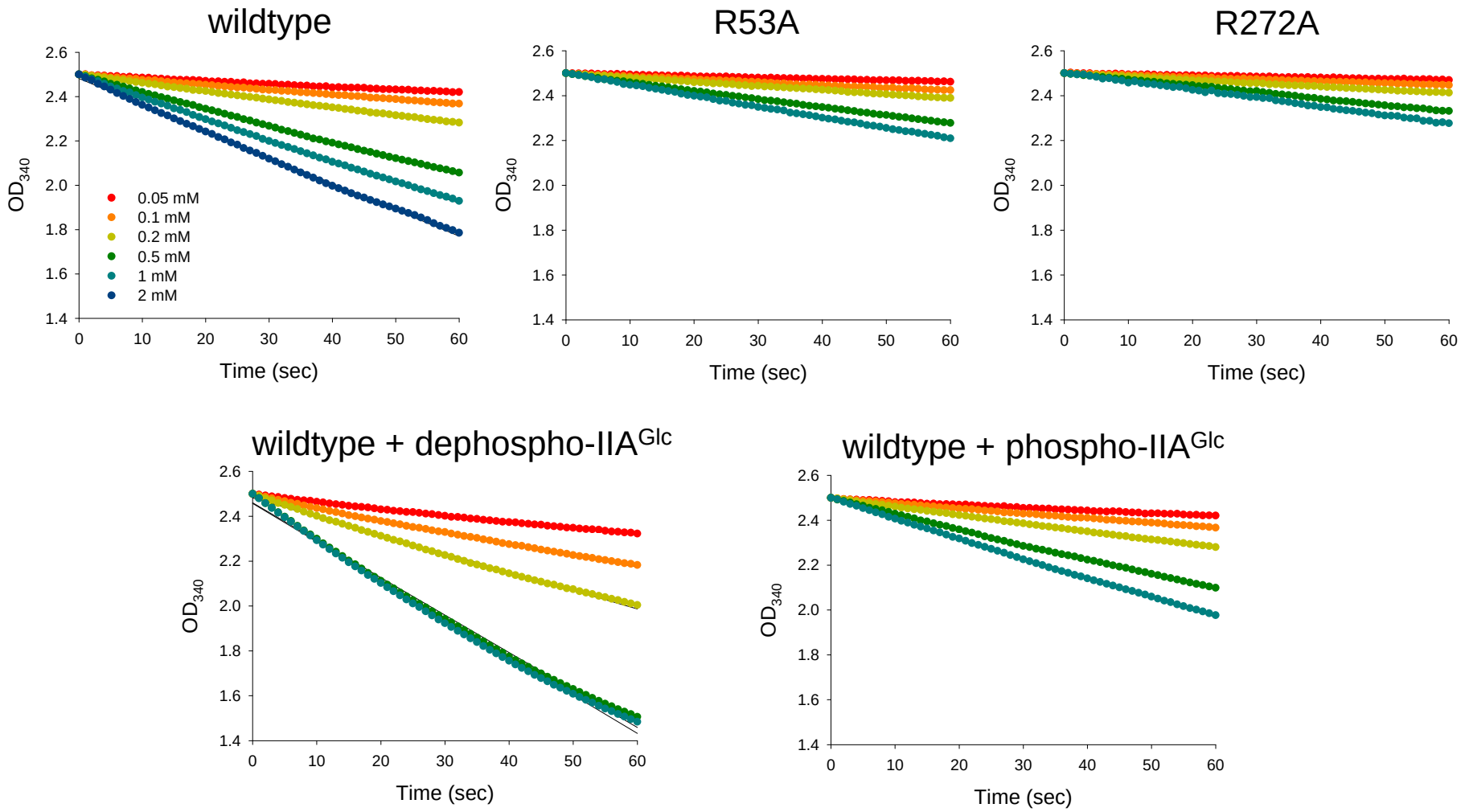
Supplementary Fig. 3. Acetaldehyde production by crude cell lysates of the wildtype *E. coli*. Various concentrations, ranging from 0.005 to 1.0 mg of protein, of crude extracts of wildtype (closed circle) and Δ*frsA* mutant (open circle) were added to a solution including 1mM pyruvate and incubated for 5 min at 37 °C. The concentrations of the produced acetaldehyde were estimated using the acetaldehyde UV-method kit (Roche) as described in Supplementary Methods.

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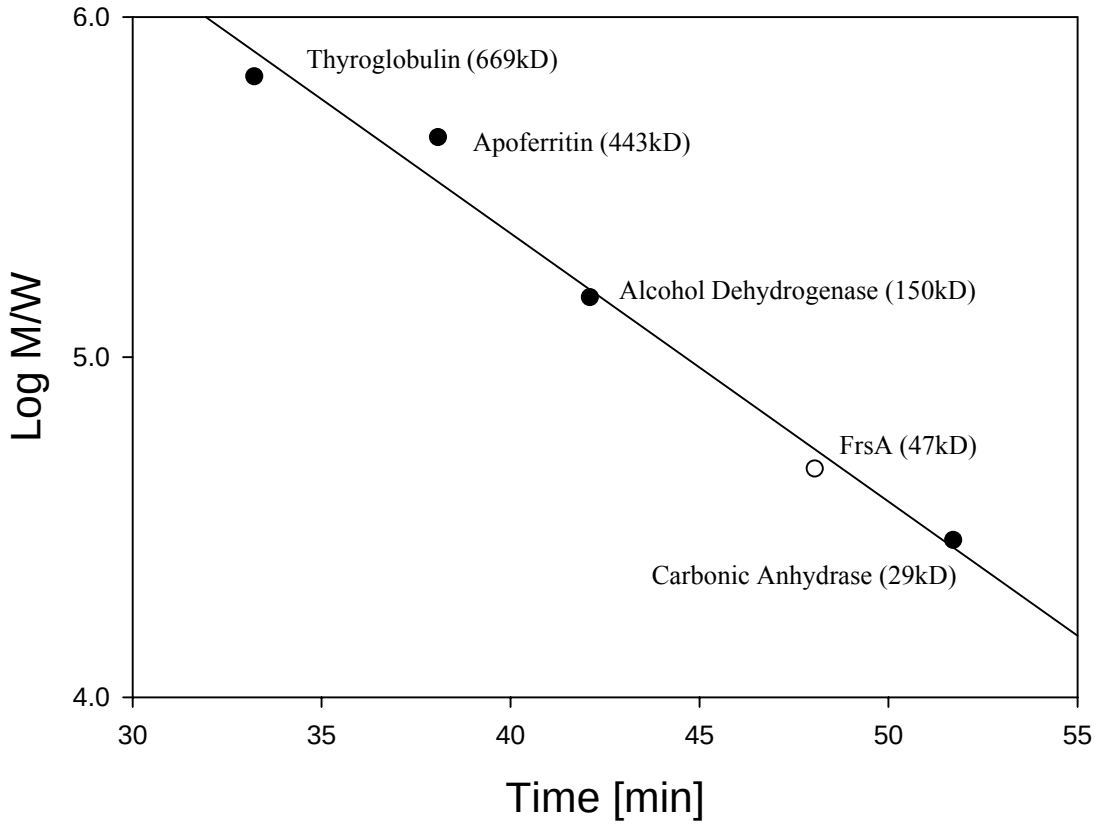
Supplementary Fig. 4. 500MHz ^{1}H-NMR spectra of FrsA reaction mixtures with α-ketobutyrate or α-ketoglutarate. The substrate specificity of FrsA was also examined with various carboxylate-containing metabolites by monitoring the production of carbon dioxide. No production of carbon dioxide was observed in reaction mixtures containing acetate, lactate, oxaloacetate, α-ketoglutarate, and α-ketobutyrate.
(a) Five millimolar α-ketobutyrate was added to the reaction mixture as described in Supplementary Fig. 2. After incubation, the reaction mixtures were analyzed by ^{1}H-NMR. The peaks of spectra indicated: KB, α-ketobutyrate; TMS, tetramethylsilane. (b**)** Five millimolar α-ketoglutarate was added to the reaction mixture as described in Supplementary Fig. 2. After incubation, the reaction mixtures were analyzed by ^{1}H-NMR.
The peaks of spectra indicated: KG, α-ketoglutarate; TMS, tetramethylsilane.

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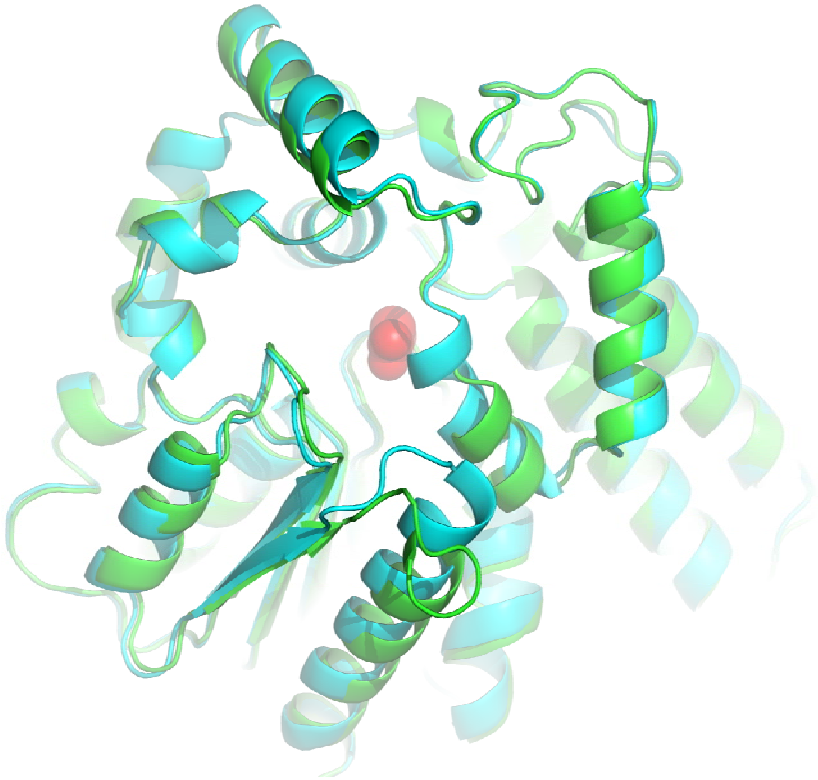
Supplementary Fig. 5. Continuous measurement of the FrsA-mediated acetaldehyde production by monitoring the NADH concentration. One nM of the recombinant FrsA proteins (47 ng of proteins in the total reaction volume of 1.0 ml) was added to 50 mM sodium phosphate solutions (pH 7.0) containing 1 mM DTT, 2 mM NADH, 3.7 U alcohol dehydrogenase (ADH), and various concentrations of pyruvate. For the reaction with the wildtype FrsA, 0.05, 0.1, 0.2, 0.5, 1.0, and 2.0 mM of pyruvate were used. In the case of the other reactions, 0.05, 0.1, 0.2, 0.5, and 1.0 mM of pyruvate were used for the kinetic analyses. Effect of IIA^{Glc} on the FrsA activity was monitored by adding 2.0 nM p-IIA^{Glc} or d-IIA^{Glc}. During one min-reaction at 37 °C, the remaining NADH in the reaction mixtures was continuously measured by spectrophotometry at 340 nm. All kinetic experiments were performed twice and the average values were plotted. The resultant kinetic parameters were presented in Table 1.

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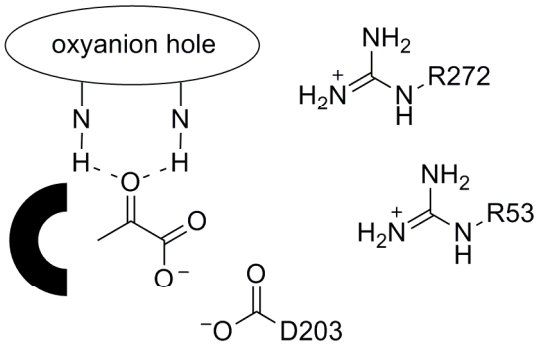
Supplementary Fig. 6. The line fitting of the elution time *versus* logarithm of the molecular weight of the size markers. Oligomeric state of FrsA was analysed by gel filtration chromatography. The proteins were eluted with a 10 mM sodium phosphate buffer (pH 8.0) containing 1mM DTT and 300 mM NaCl. Predicted molecular mass of FrsA is 47 kDa, indicating that FrsA exists as a monomer in solution.

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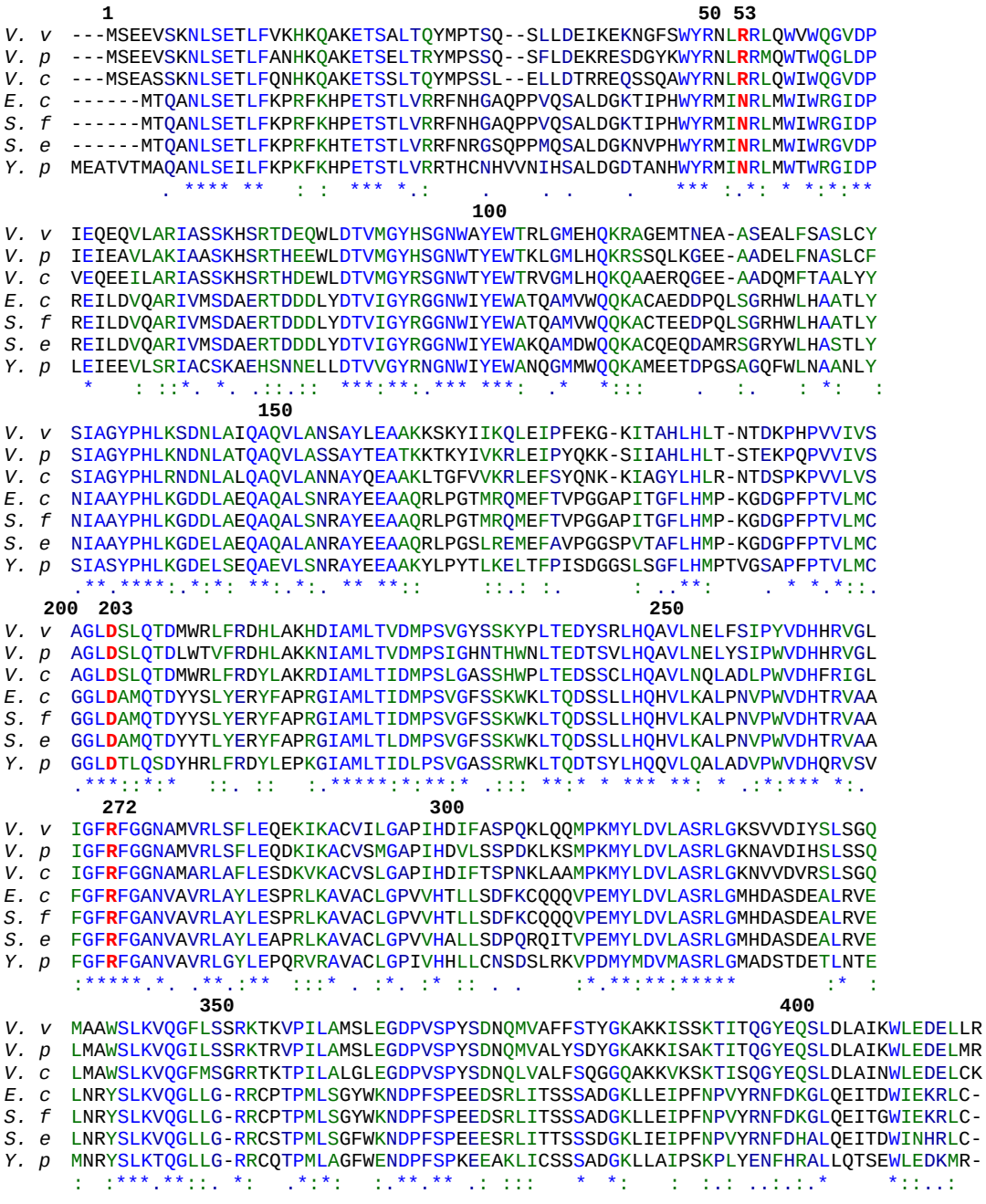
Supplementary Fig. 7. Ribbon diagram of superposed FrsA-I and -II. Asp203 is represented by red spheres.

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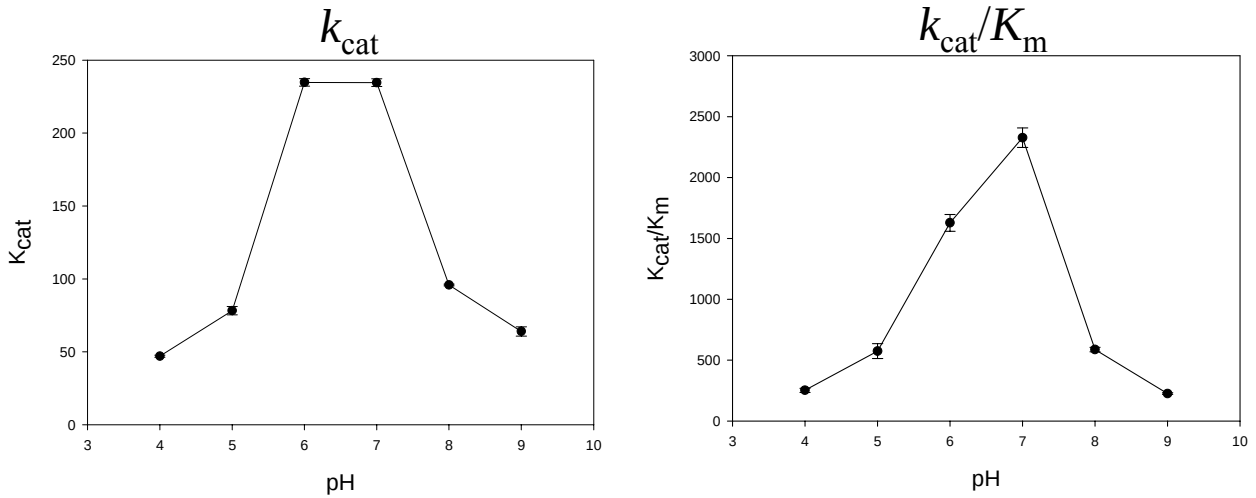
Supplementary Fig.8. Schematic drawing to predict interactions of pyruvate with FrsA. The black pocket represents the hydrophobic pocket in the active site. Black dots indicate interactions between the carbonyl oxygen and two -NH groups in the oxyanion hole.

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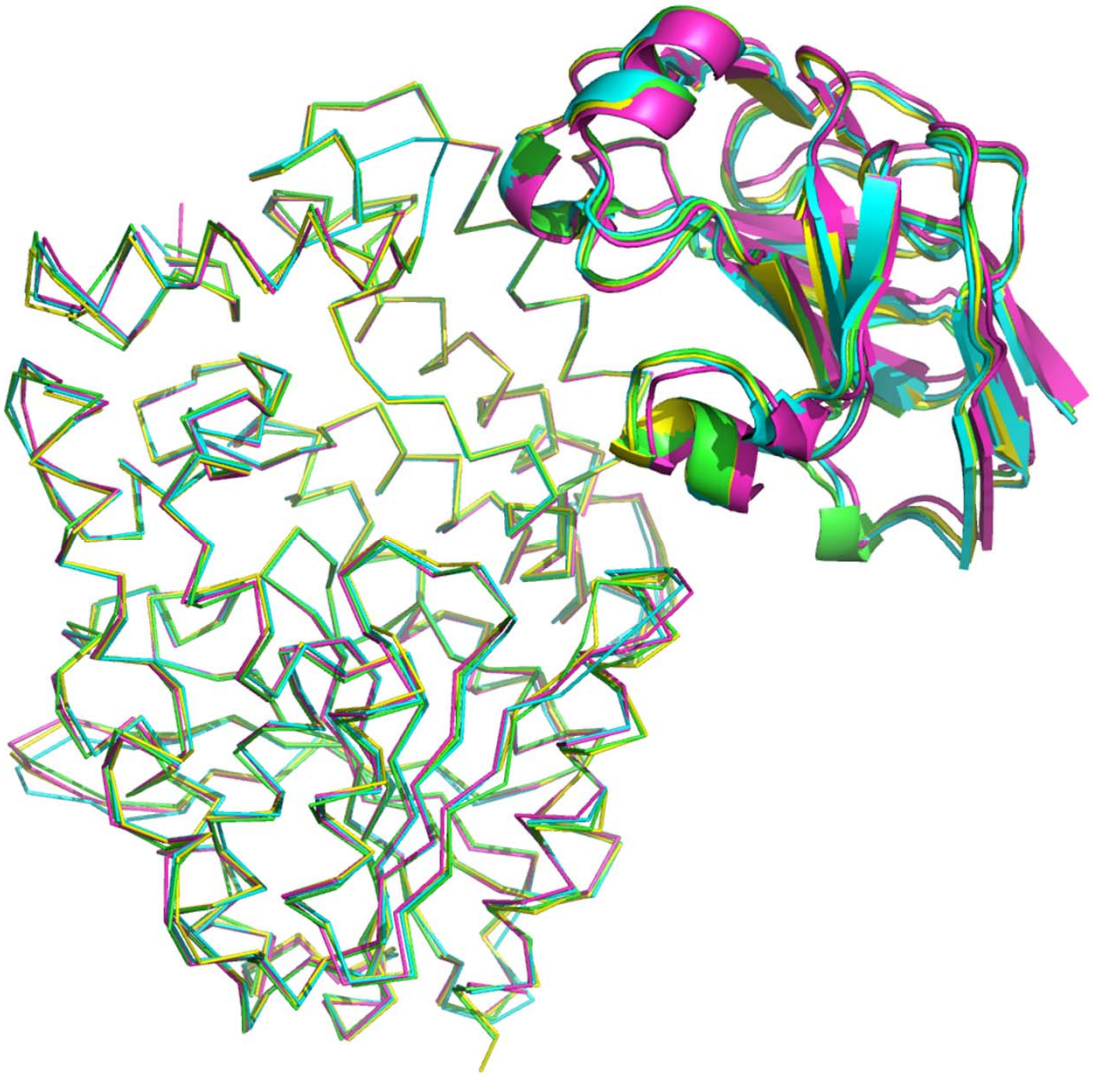
Supplementary Fig. 9. Amino acid sequence alignment of the *V. vulnificus* FrsA with its orthologs. *V. vulnificus* (*V. v*) FrsA sequence was aligned with the FrsA orthologs deposited to the GenBank using the program ClustalW. The residues of identical and similar amino acids are marked with the blue asterisks and the green dots, respectively. Three catalytic residues (R53, D203, and R272) are highlighted by red color. Amino acid sequences are from the NCBI Database: *V. v* (*Vibrio vulnificus*; NP_759330.1); *V. p* (*Vibrio parahaemolyticus*; EED26191); *V. c* (*Vibrio cholerae*; NP_231907); *E. c* (*Escherichia coli*; AP_000892); *S. f* (*Shigella flexneri*; NP_706239); *S. e* (*Salmonella enterica*; YP_001571692); and *Y. p* (*Yersinia pestis*; AAM84545).

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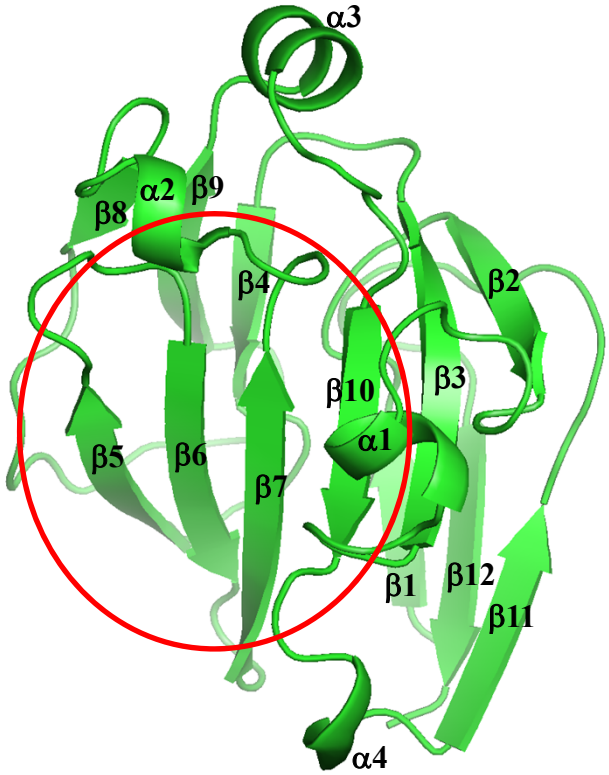
Supplementary Fig. 10. The pH dependence of FrsA-catalyzed reaction. The recombinant FrsA was added to the reaction mixtures titrated to various pH, ranging from 4 to 9, and then its enzymatic kinetic parameters were estimated using the alcohol dehydrogeanse (ADH) assay. For estimating acetaldehyde produced from pyruvate, 2 mM NADH and 3.7 U yeast alcohol dehydrogenase (ADH) (Sigma Co.) were included in the FrsA reaction mixture. Reaction was initiated by adding FrsA (1 nM) and reaction mixtures were incubated at 37 ^{o}C for 5 min. After terminating the reactions by adding 50 mM potassium hydrogen phthalate buffer (pH 3.0), the remained NADH in the reaction mixtures were determined by spectrophotometry at 340 nm. The values of k_{cat} and k_{cat}/K_m for the conversion of pyruvate to acetaldehyde were determined by the standard curves of ADH reaction performed under the corresponding pH condition.

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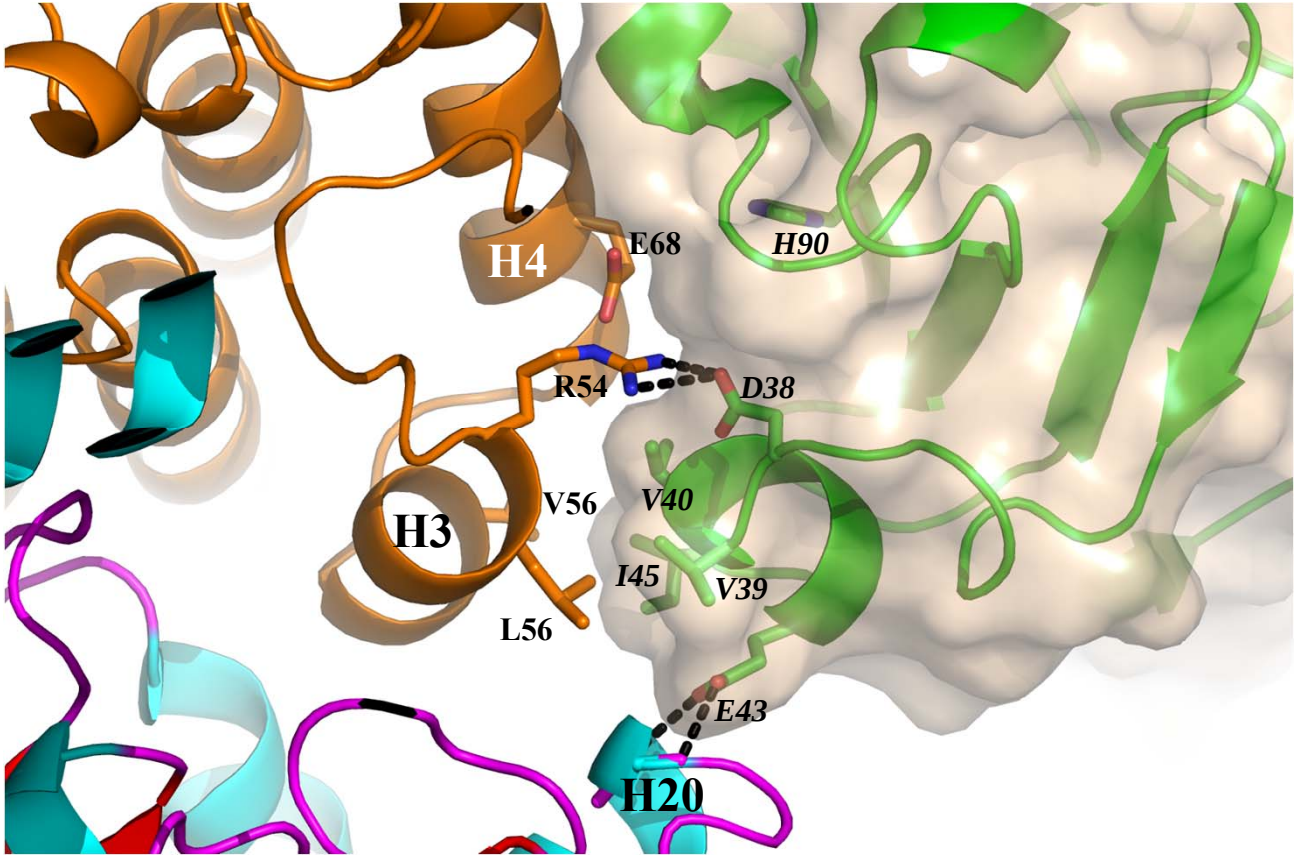
Supplementary Fig. 11. Cα tracing of superposed FrsA/d-IIA^{Glc} complexes in the asymmetric unit. FrsA and d-IIA^{Glc} form a complex at a molar stoichiometry of 1:1. The FrsA/d-IIA^{Glc} complex yielded *C*-centered monochrinic crystals with four complexes in an asymmetric unit whose structures are well superposed with root mean square deviation of ~0.4 Å for all Cα atoms. For this figure, the four complexes are differently colored. For clarity, d-IIA^{Glc} is represented by cartoon.

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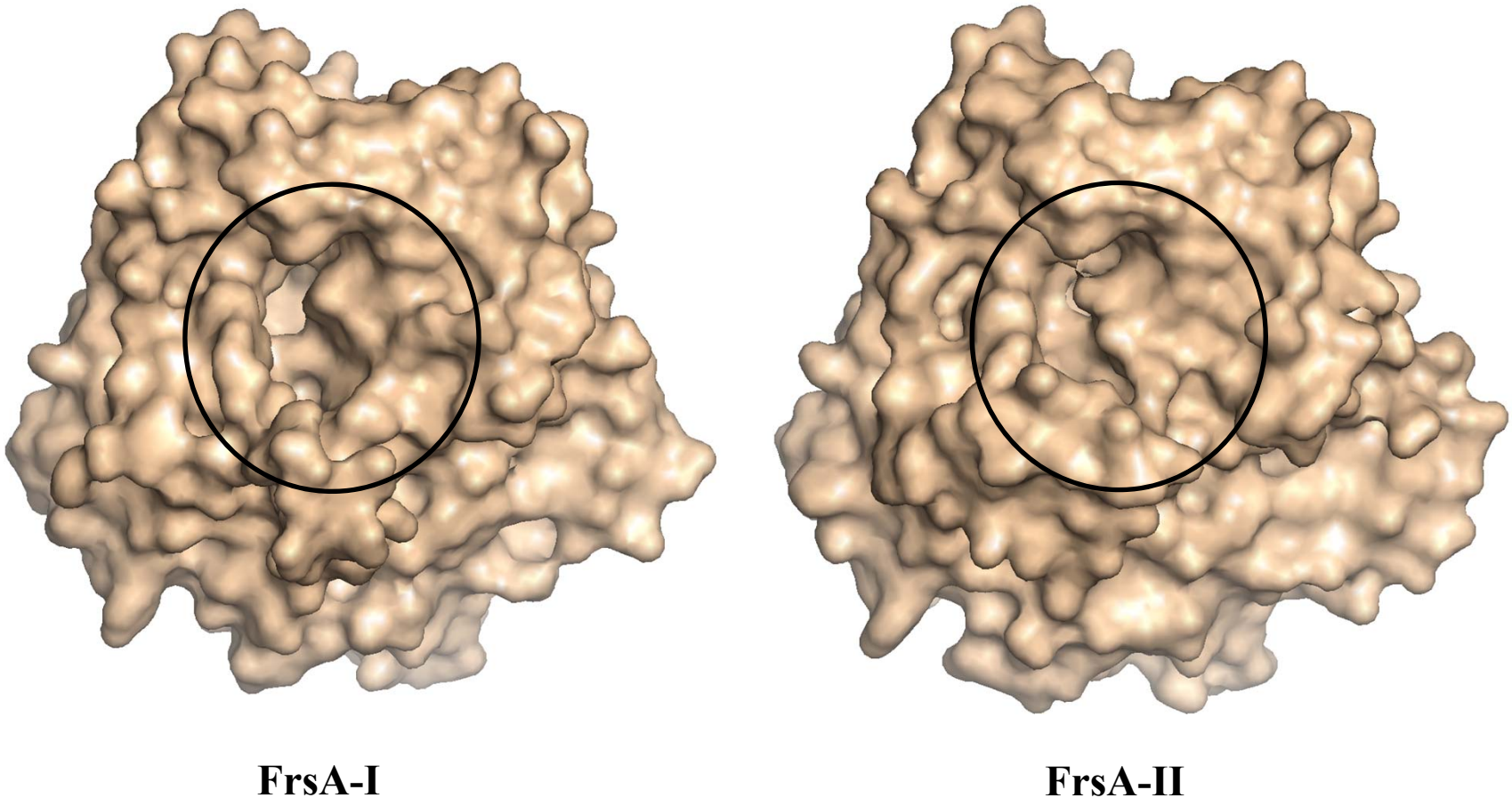
Supplementary Fig.12. Overall structure of IIA^{Glc}. Ribbon diagram of IIA^{Glc} shown with secondary structure labeled. d-IIA^{Glc} contains one six-stranded antiparallel β-sheet (strands 2, 3, 5, 6, 7, and 10) and two three-stranded antiparallel β-sheets (strands 1, 11, and 12; strands 4, 8, and 9). The two three-stranded sheets pack against the six-stranded sheet to form a β sandwich with four helices. The concaved surface of the six-stranded sheet in d-IIA^{Glc}, which is represented by a red circle, binds to FrsA, excluding 1,691 Å^2 solvent accessible surface area.

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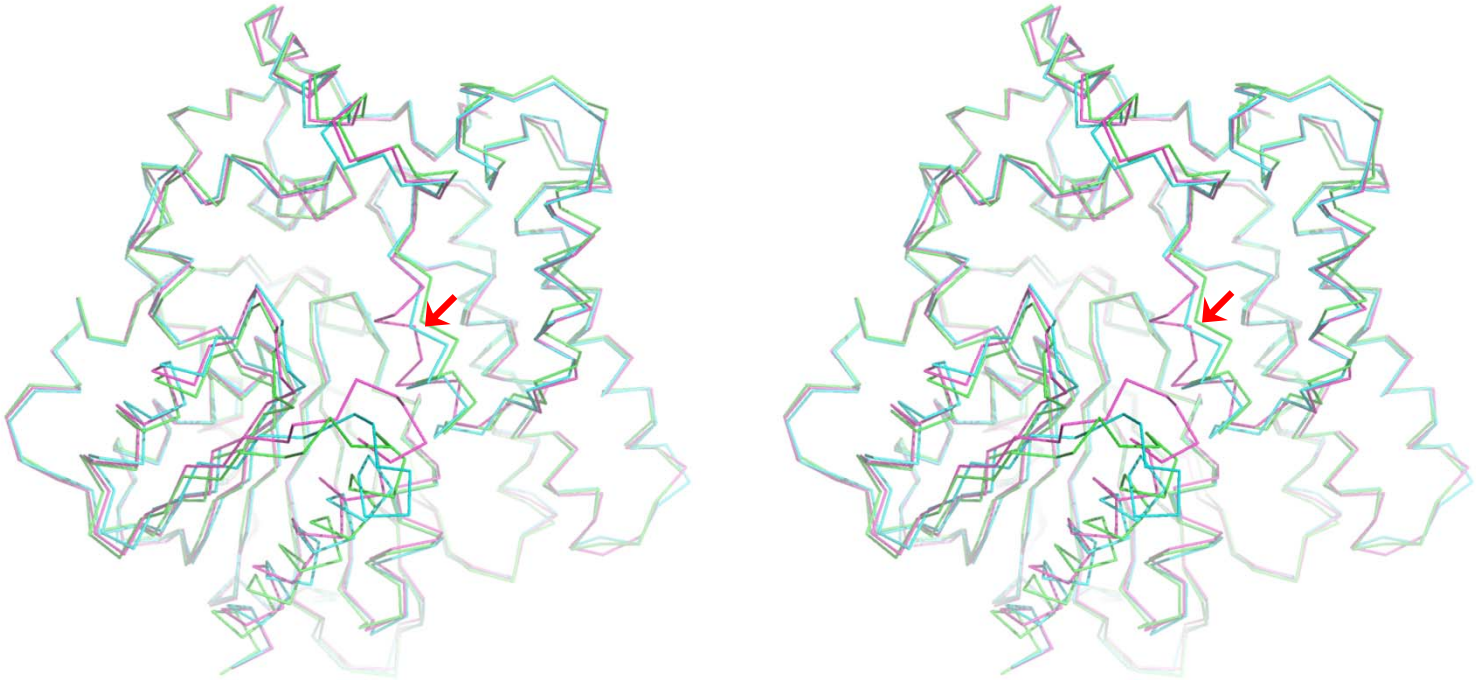
Supplementary Fig.13. Close-up view of the binding interface. Color scheme is identical to that shown in Fig. 2c. Dotted lines in black represent possible polar interactions. The binding interface can be divided into three sectors: H3, H4, and H20 sectors depending on the interacting helix of FrsA. In the H3 sector, Leu56 and Val58 of FrsA make hydrophobic interactions with Val39, Val40, and Ile45 of d-IIA^{Glc}, and Arg54 of FrsA makes an ion pair with Asp38 of d-IIA^{Glc}. In the H4 sector, the H4 helix of FrsA makes extensive hydrophobic and hydrophilic contact with d-IIA^{Glc} along the helix. It is worth noting that Glu68 of FrsA is located in the vicinity of the phosphorylation site (His90) of d-IIA^{Glc}. In the H20 sector, the backbone -NH groups at the amino terminus of H20 make hydrogen bonds with the side-chain carboxylate group of Glu43 in d-IIA^{Glc}.

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Supplementary Fig.14. Surface representations of FrsA-I and II. Black circles indicate the active site. Surface representations revealed that FrsA-I and -II have an open and a blocked active sites, respectively, indicating that FrsA probably has dual active site conformation in solution. Considering that pyruvate cannot access to the blocked active site, it can be safely assumed that FrsA-I and -II have the active and inactive conformations, respectively.

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Supplementary Fig.15. Stereo view of the superimposed Cα tracing of FrsA-I (cyan), FrsA-II (magenta), and FrsA in the complex (green).
The disparity in the active site between FrsA-I and -II can be attributed to the conformational difference of helix H3 pointed out by a red arrow. The H3 helix in FrsA-II is longer than that in FrsA-I due to the helical conformation of residues 52-53 which are in a loop in FrsA-I. The helical region of residues 52-53 in FrsA-II extends into the active site, thereby blocking the entrance of the active site.
The structure of FrsA in the complex with the open active site resembles that of FrsA-I, suggesting that the binding of d-IIA^{Glc} to FrsA keeps the active site open.

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Supplementary Table 1 Lethality of wildtype and Δ*frsA* mutant *V. vulnificus* strains to mice

Strain	Genotype	No. of dead mice / No. of total mice					LD_{50}
		No. of injected bacteria					
		7.5×10^1	7.5×10^2	7.5×10^3	7.5×10^4	7.5×10^5	(CFU)
MO6-24/O	wildtype	0:6	1:6	4:6	6:6	6:6	3.2×10^3
SM201	Δ*frsA*	0:6	0:6	0:6	4:6	5:6	5.7×10^4

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Supplementary Table 2 Data collection, phasing and refinement statistics for MAD (SeMet) structures

	FrsA-Native		FrsA-MAD	
Data collection				
Space group			$P2_1$	
Cell dimensions				
a, *b*, *c* (Å)		60.83	88.20	79.63
α, β, γ (°)		90	106.62	90
		Peak	*Inflection*	*Remote*
Wavelength	0.98000	097966	0.97990	0.98778
Resolution (Å)	50-2.18(2.22-2.18)*	50-3.0	50-3.0	50-3.0
R_{sym} or R_{merge}	6.9(22.1)	7.7(22.5)	7.2(22.6)	5.8(19.6)
$I / \sigma I$	26.2(4.56)	18.5(2.88)	17.9(2.78)	18.9(3.04)
Completeness (%)	98.3(84.4)	96.1(89.9)	96.1(89.8)	96(90.1)
Redundancy	4.7(3.5)	3.3(2.5)	3.3(2.5)	3.3(2.5)
Refinement				
Resolution (Å)	48.63-2.2			
No. reflections				
R_{work} / R_{free}	18.5/22.9			
No. atoms	39288			
Protein	6136			
Sufate	5			
Ethylene glycol	10			
Water	304			
B-factors				
Protein	27.38			
Sulfate	87.22			
Ethylene glycol	45.45			
Water	32.22			
R.m.s deviations				
Bond lengths (Å)	0.008			
Bond angles (°)	1.286			

*Highest-resolution shell is shown in parentheses.

Supplementary Table 3 Data collection and refinement statistics

	FrsA-IIA^{Glc}
Data collection	
Space group	*C*2
Cell dimensions	
a, *b*, *c* (Å)	322.87, 62.07, 126.36
α, β, γ (°)	90, 110.80, 90
Resolution (Å)	50-2.19(2.23-2.19)*
R_{sym} or R_{merge}	7.7(35.9)
$I / \sigma I$	14.69(1.75)
Completeness (%)	98.5(95.1)
Redundancy	4.4(3.1)
Refinement	
Resolution (Å)	49.37-2.2
No. reflections	108181
R_{work} / R_{free}	20.1/24.8
No. atoms	
Protein	16727
Water	933
B-factors	
Protein	36.59
Water	42.34
R.m.s. deviations	
Bond lengths (Å)	0.009
Bond angles (°)	1.494

*Highest-resolution shell is shown in parentheses.

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